Pradyunn Kale

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EDUCATION

Purdue University West Lafayette, IN
Bachelor of Science in Electrical Engineering, Cumulative GPA: 3.79/4.00 August 2025 – May 2028
Current Coursework: ASIC Design, Circuit Analysis II
Organizations: SCalable Asymmetric Lifecycle Engagement (SCALE)

EXPERIENCE

IEEE Electron Devices Society *September 2026 – Present*
Mixed-Signal IC Design West Lafayette, IN

- Designing an 8 GHz quadrature-output ADPLL for a 32 Gb/s PAM4 optical SerDes, implementing a dual-mode architecture for coarse frequency acquisition and fine phase tracking with threshold-based mode switching.
- Developing a hybrid TDC with coarse frequency counter and fine gated ring oscillator, SAR/PI digital loop filter, programmable fractional-N divider, and Class-F quadrature DCO with coarse/fine frequency tuning.

Purdue UAS Research & Test Facility (SURF 2026) *March 2026 – September 2026*
Undergraduate Researcher West Lafayette, IN

- Built Rumoca's Rust GALEC admissibility layer for eFMI control models and optimized embedded C generation.
- Applied the Rust-based Rumoca compiler toolchain to fixed-wing autopilot control algorithms, including the Total Energy Control System (TECS) for aircraft energy management and PID controllers for attitude tracking.
- Integrated compiler-generated embedded C control logic into a real-time flight controller, configuring execution task priorities and control subsystem interfaces to meet the flight controller's real-time execution constraints.

Purdue Space Program – Liquid Propulsion *July 2025 – September 2026*
Flight Dynamics & Controls Lead / Avionics Engineer West Lafayette, IN

- Designed preliminary flight computer estimation with sensor data capture and INS/Kalman filter interfaces.
- Collaborated on sensor error and noise models in a MATLAB 6-DoF rocket simulation for Kalman filter tuning.

PROJECTS

8x8 Floating Point Systolic Array Accelerator *November 2026 – December 2026*

- A weight-stationary systolic array in SystemVerilog with 64 multiply-accumulate (MAC) processing elements, FP16 internal arithmetic, E4M3 input/output conversion, and streamed weight loading through a 64-bit interface.
- Selectable activation function datapaths support ReLU, Leaky ReLU, binary, and identity functions in an AHB-connected accelerator that operates at 100 MHz with inference latency of 55 clock cycles or fewer.
- Built a C-based golden model, imported it into SystemVerilog testbenches through DPI, and exhaustively compared standalone FP16 multiplier and adder outputs against the reference across supported input combinations.
- Collaborating with two teammates to integrate the systolic array and selectable activation datapaths with an AHB subordinate interface, coordinating connections to the controller and SRAM data buffer subsystem.

TECHNICAL SKILLS

Programming: C, SystemVerilog, Python, Bash, Rust, MATLAB, Simulink.

Embedded Systems: FPGA, STM32, Zephyr/FreeRTOS, SPI/I^2C/UART, Ethernet, JTAG/SWD, Oscilloscope.

Tools: Git, Linux, Nix, Siemens QuestaSim, Synopsys Design Compiler, Cadence Innovus, Altium, KiCAD, LTSpice.